ANSWER KEY

빠른정답

01 $\dfrac{6(x^2-3x+4)}{(x-5)(x-2)(x-1)(x+2)}$	**02** $\dfrac{20}{29}$	**03** $\dfrac{x+2}{x-1}$	
04 ②	**05** ③	**06** ④	**07** 40
08 4	**09** ①	**10** ③	**11** 6
12 ②	**13** ②	**14** 4	**15** $-\dfrac{64x}{x^4+4x^2+16}$
16 8	**17** ⑤	**18** ③	**19** 10
20 ②	**21** ②	**22** 18	**23** $112\sqrt{3}$
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오답 메모 · 다시 볼 문항

해설 SOLUTIONS

01 정답 $\frac{6(x^2-3x+4)}{(x-5)(x-2)(x-1)(x+2)}$

$$\frac{1}{x-5}+\frac{1}{x-1}-\frac{1}{x-2}-\frac{1}{x+2}$$
$$=\left(\frac{1}{x-5}-\frac{1}{x-2}\right)+\left(\frac{1}{x-1}-\frac{1}{x+2}\right)$$
$$=\frac{x-2-(x-5)}{(x-5)(x-2)}+\frac{x+2-(x-1)}{(x-1)(x+2)}$$
$$=\frac{3}{(x-5)(x-2)}+\frac{3}{(x-1)(x+2)}$$
$$=\frac{3(x-1)(x+2)+3(x-5)(x-2)}{(x-5)(x-2)(x-1)(x+2)}$$
$$=\frac{3x^2+3x-6+3x^2-21x+30}{(x-5)(x-2)(x-1)(x+2)}$$
$$=\frac{6x^2-18x+24}{(x-5)(x-2)(x-1)(x+2)}$$
$$=\frac{6(x^2-3x+4)}{(x-5)(x-2)(x-1)(x+2)}$$

02 정답 $\frac{20}{29}$

$$\frac{a+b}{a-b}-\frac{a-b}{a+b}=\frac{(a+b)^2-(a-b)^2}{(a-b)(a+b)}=\frac{4ab}{(a-b)(a+b)}$$
$$\frac{a+b}{a-b}+\frac{a-b}{a+b}=\frac{(a+b)^2+(a-b)^2}{(a-b)(a+b)}=\frac{2(a^2+b^2)}{(a-b)(a+b)}$$
$$\therefore\left(\frac{a+b}{a-b}-\frac{a-b}{a+b}\right)\div\left(\frac{a+b}{a-b}+\frac{a-b}{a+b}\right)$$
$$=\frac{4ab}{(a-b)(a+b)}\times\frac{(a-b)(a+b)}{2(a^2+b^2)}$$
$$=\frac{2ab}{a^2+b^2}=\frac{20}{29}$$

03 정답 $\frac{x+2}{x-1}$

$$A\div\frac{x^3+8}{x^4+4x^2+16}\times\frac{x^2-x}{x^3-8}$$
$$=A\times\frac{(x^2-2x+4)(x^2+2x+4)}{(x+2)(x^2-2x+4)}$$
$$\times\frac{x(x-1)}{(x-2)(x^2+2x+4)}$$
$$=A\times\frac{x(x-1)}{(x+2)(x-2)}$$

따라서 $A\times\frac{x(x-1)}{(x+2)(x-2)}=\frac{x}{x-2}$ 이므로

$$A=\frac{x}{x-2}\times\frac{(x+2)(x-2)}{x(x-1)}=\frac{x+2}{x-1}$$

04 정답 ②

$$\frac{1}{a-1}-\frac{1}{a+1}-\frac{2}{a^2+1}-\frac{4}{a^4+1}$$
$$=\frac{(a+1)-(a-1)}{(a-1)(a+1)}-\frac{2}{a^2+1}-\frac{4}{a^4+1}$$
$$=\frac{2}{a^2-1}-\frac{2}{a^2+1}-\frac{4}{a^4+1}$$
$$=\frac{2(a^2+1)-2(a^2-1)}{(a^2-1)(a^2+1)}-\frac{4}{a^4+1}$$
$$=\frac{4}{a^4-1}-\frac{4}{a^4+1}$$
$$=\frac{4(a^4+1)-4(a^4-1)}{a^8-1}$$
$$=\frac{2\times4}{a^8-1}=\frac{2^3}{a^8-1}$$

$p=3,\ q=8$ 이므로 $p+q=11$ 이다.

05 정답 ③

$\frac{3x+23}{x^2-3x-4}=\frac{a}{x+1}+\frac{b}{x-4}$ 에서

$$\frac{3x+23}{x^2-3x-4}=\frac{a(x-4)+b(x+1)}{(x+1)(x-4)}$$
$$\frac{3x+23}{x^2-3x-4}=\frac{(a+b)x-4a+b}{x^2-3x-4}$$

이때 주어진 등식은 $x\neq-1,\ x\neq4$ 인 모든 실수 x 에 대하여 성립하므로

$a+b=3,\ -4a+b=23$

$\therefore\ a=-4,\ b=7$

$\therefore\ ab=-28$

06 정답 ④

$12x^2-2ax+b\neq0$ 인 모든 실수 x 에 대하여 유리식 $\frac{3x^2+2x+a}{12x^2-2ax+b}$ 의 값이 항상 일정하므로

$\frac{\alpha\times A(x)}{\beta\times A(x)}$ ($\alpha,\ \beta$ 는 상수이고 $\beta\neq0,\ A(x)$ 는 x 에 대한 다항식) 꼴이어야 한다.

즉, $\frac{3x^2+2x+a}{12x^2-2ax+b}=\frac{3x^2+2x+a}{4\left(3x^2-\frac{a}{2}x+\frac{b}{4}\right)}$ 이므로

$-\frac{a}{2}=2,\ \frac{b}{4}=a$

$\therefore\ a=-4,\ b=-16$

$\therefore\ a-b=12$

해설 SOLUTIONS

07 정답 40

$$\frac{a_1}{x-2}+\frac{a_2}{(x-2)^2}+\cdots+\frac{a_5}{(x-2)^5}=\frac{x^4-20}{(x-2)^5}$$

양변에 $(x-2)^5$을 곱하면

$a_1(x-2)^4+a_2(x-2)^3+\cdots+a_4(x-2)+a_5=x^4-20$ ····· ㉠

㉠의 양변에 $x=1$을 대입하면

$a_1-a_2+a_3-a_4+a_5=-19$ ····· ㉡

㉠의 양변에 $x=3$을 대입하면

$a_1+a_2+a_3+a_4+a_5=61$ ····· ㉢

㉢−㉡을 계산하면

$2(a_2+a_4)=80$

$\therefore\ a_2+a_4=40$

08 정답 4

$$\frac{x}{x-2}-\frac{x+1}{x-1}-\frac{x+2}{x}+\frac{x+3}{x+1}$$

$$=\frac{(x-2)+2}{x-2}-\frac{(x-1)+2}{x-1}-\frac{x+2}{x}+\frac{(x+1)+2}{x+1}$$

$$=\left(1+\frac{2}{x-2}\right)-\left(1+\frac{2}{x-1}\right)-\left(1+\frac{2}{x}\right)+\left(1+\frac{2}{x+1}\right)$$

$$=\frac{2}{x-2}-\frac{2}{x-1}-\frac{2}{x}+\frac{2}{x+1}$$

$$=\frac{2}{(x-2)(x-1)}-\frac{2}{x(x+1)}$$

$$=\frac{2\{x(x+1)-(x-2)(x-1)\}}{x(x-2)(x-1)(x+1)}$$

$$=\frac{8x-4}{x(x-2)(x-1)(x+1)}$$

즉,

$$\frac{8x-4}{x(x-2)(x-1)(x+1)}=\frac{ax+b}{x(x-2)(x-1)(x+1)}$$

가 x에 대한 항등식이므로 $a=8,\ b=-4$

$\therefore\ a+b=4$

09 정답 ①

$$\frac{3x^2-3x+5}{x-1}-\frac{3x^2+9x-10}{x+3}$$

$$=\frac{3x(x-1)+5}{x-1}-\frac{3x(x+3)-10}{x+3}$$

$$=3x+\frac{5}{x-1}-\left(3x-\frac{10}{x+3}\right)$$

$$=\frac{5}{x-1}+\frac{10}{x+3}$$

$$=\frac{5(x+3)+10(x-1)}{(x-1)(x+3)}$$

$$=\frac{15x+5}{(x-1)(x+3)}$$

따라서 $f(x)=15x+5$이므로 $f(1)=20$이다.

10 정답 ③

$$\frac{1}{4}+\frac{1}{28}+\frac{1}{70}+\cdots+\frac{1}{550}$$

$$=\frac{1}{1\times4}+\frac{1}{4\times7}+\frac{1}{7\times10}+\cdots+\frac{1}{22\times25}$$

$$=\frac{1}{3}\left\{\left(\frac{1}{1}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{7}\right)+\cdots+\left(\frac{1}{22}-\frac{1}{25}\right)\right\}$$

$$=\frac{1}{3}\left(1-\frac{1}{25}\right)=\frac{8}{25}$$

11 정답 6

$$\frac{1}{x^2+9x+18}+\frac{1}{x^2+3x}+\frac{1}{x^2-3x}$$

$$=\frac{1}{(x+6)(x+3)}+\frac{1}{x(x+3)}+\frac{1}{x(x-3)}$$

$$=\frac{1}{3}\left\{\left(\frac{1}{x+3}-\frac{1}{x+6}\right)+\left(\frac{1}{x}-\frac{1}{x+3}\right)+\left(\frac{1}{x-3}-\frac{1}{x}\right)\right\}$$

$$=\frac{1}{3}\left(\frac{1}{x-3}-\frac{1}{x+6}\right)$$

$$=\frac{3}{(x+6)(x-3)}$$

따라서 $\frac{3}{(x+6)(x-3)}=\frac{1}{12}$이므로 $(x+6)(x-3)=36$

$x^2+3x-54=0,\ (x+9)(x-6)=0$

$\therefore\ x=6\ (\because\ x>0)$

해설 SOLUTIONS

12 정답 ②

$$\frac{4}{(x+3)(x+7)}+\frac{5}{(x+7)(x+12)}+\frac{6}{(x+12)(x+18)}+\cdots+\frac{9}{(x+33)(x+42)}$$

$$=\left(\frac{1}{x+3}-\frac{1}{x+7}\right)+\left(\frac{1}{x+7}-\frac{1}{x+12}\right)+\left(\frac{1}{x+12}-\frac{1}{x+18}\right)+\cdots+\left(\frac{1}{x+33}-\frac{1}{x+42}\right)$$

$$=\frac{1}{x+3}-\frac{1}{x+42}$$

$$=\frac{1}{6}-\frac{1}{45}=\frac{13}{90}$$

13 정답 ②

$$\frac{1}{1+\dfrac{1}{1+\dfrac{1}{x}}}=\frac{1}{1+\dfrac{1}{\dfrac{x+1}{x}}}$$

$$=\frac{1}{1+\dfrac{x}{x+1}}$$

$$=\frac{1}{\dfrac{(x+1)+x}{x+1}}$$

$$=\frac{x+1}{2x+1}$$

즉, $\dfrac{x+1}{2x+1}=\dfrac{ax+b}{cx+1}$ 는 x 에 대한 항등식이므로

$a=1,\ b=1,\ c=2$

$\therefore\ a+2b+3c=1+2+6=9$

14 정답 4

$$f(x)=6-\frac{2+\dfrac{4}{x-2}}{2-\dfrac{4}{x+2}}$$

$$=6-\frac{\dfrac{2x}{x-2}}{\dfrac{2x}{x+2}}$$

$$=6-\frac{x+2}{x-2}=\frac{5x-14}{x-2}$$

따라서 $f(a)=3$ 에서 $\dfrac{5a-14}{a-2}=3$

$5a-14=3a-6\ \therefore\ a=4$

15 정답 $-\dfrac{64x}{x^4+4x^2+16}$

$x^3-8=(x^3+8)-16=(x+2)(x^2-2x+4)-16,$

$x^3+8=(x^3-8)+16=(x-2)(x^2+2x+4)+16$

이므로

$$\frac{x^3-8}{x^2-2x+4}+\frac{x^3+8}{x^2+2x+4}-2x$$

$$=\frac{(x+2)(x^2-2x+4)-16}{x^2-2x+4}$$

$$+\frac{(x-2)(x^2+2x+4)+16}{x^2+2x+4}-2x$$

$$=\left(x+2-\frac{16}{x^2-2x+4}\right)$$

$$+\left(x-2+\frac{16}{x^2+2x+4}\right)-2x$$

$$=\frac{16}{x^2+2x+4}-\frac{16}{x^2-2x+4}$$

$$=-\frac{64x}{x^4+4x^2+16}$$

16 정답 8

$$\frac{12}{17}=1-\frac{5}{17}=1-\frac{1}{\dfrac{17}{5}}=1-\frac{1}{3+\dfrac{2}{5}}$$

$$=1-\frac{1}{3+\dfrac{1}{\dfrac{5}{2}}}=1-\frac{1}{3+\dfrac{1}{2+\dfrac{1}{2}}}$$

따라서 $a=1,\ b=3,\ c=2,\ d=2$ 이므로

$a+b+c+d=8$ 이다.

17 정답 ⑤

$$\frac{\dfrac{1}{n+2}-\dfrac{1}{n+9}}{\dfrac{1}{n+9}-\dfrac{1}{n+16}}=\frac{\dfrac{7}{(n+2)(n+9)}}{\dfrac{7}{(n+9)(n+16)}}$$

$$=\frac{n+16}{n+2}\ (\because\ n\neq -9)$$

$$=1+\frac{14}{n+2}$$

자연수 n 에 대하여 $1+\dfrac{14}{n+2}$ 가 자연수이려면

$n+2$ 는 14 의 양의 약수이어야 한다.

따라서 모든 자연수 n 의 값의 합은

$5+12=17$

해설 SOLUTIONS

18 정답 ③

$x^2-5x-1=0$ 에서 $x\neq 0$ 이므로 양변을 x 로 나누면

$x-\dfrac{1}{x}=5$

$\therefore\ 2x^2-3x+6+\dfrac{3}{x}+\dfrac{2}{x^2}$

$=2\left(x^2+\dfrac{1}{x^2}\right)-3\left(x-\dfrac{1}{x}\right)+6$

$=2\left\{\left(x-\dfrac{1}{x}\right)^2+2\right\}-3\left(x-\dfrac{1}{x}\right)+6$

$=2\times 27-3\times 5+6$

$=45$

19 정답 10

$\dfrac{x+y}{4}=y+z=\dfrac{z+x}{7}=k\ (k\neq 0)$ 이라 하면

$x+y=4k,\ y+z=k,\ z+x=7k$ ····· ㉠

위의 세 식을 더하면

$2(x+y+z)=12k\ \therefore\ x+y+z=6k$ ····· ㉡

㉠, ㉡에 의하여

$x=5k,\ y=-k,\ z=2k$

$\therefore\ \dfrac{x^2+y^2+z^2}{xy+yz+zx}$

$=\dfrac{(5k)^2+(-k)^2+(2k)^2}{5k\times(-k)+(-k)\times 2k+2k\times 5k}=\dfrac{30k^2}{3k^2}=10$

20 정답 ②

$a+b-c=0$ 에서

$a+b=c,\ b-c=-a,\ a-c=-b$

이므로

$\dfrac{a^2+b^2}{ab}-\dfrac{b^2+c^2}{bc}-\dfrac{a^2+c^2}{ac}=\dfrac{a}{b}+\dfrac{b}{a}-\dfrac{b}{c}-\dfrac{c}{b}-\dfrac{a}{c}-\dfrac{c}{a}$

$=\dfrac{b-c}{a}+\dfrac{a-c}{b}-\dfrac{a+b}{c}$

$=\dfrac{-a}{a}+\dfrac{-b}{b}-\dfrac{c}{c}$

$=-1+(-1)+(-1)=-3$

21 정답 ②

$x+y-z=0$ ····· ㉠

$2x-y+2z=0$ ····· ㉡

㉠+㉡을 하면

$3x+z=0\ \therefore\ z=-3x$

$z=-3x$ 를 ㉠에 대입하면

$x+y-(-3x)=0\ \therefore\ y=-4x$

$\therefore\ \dfrac{z^2+xy}{xy+yz-zx}$

$=\dfrac{(-3x)^2+x\times(-4x)}{x\times(-4x)+(-4x)\times(-3x)-(-3x)\times x}$

$=\dfrac{5x^2}{11x^2}=\dfrac{5}{11}$

22 정답 18

$\left(x+\dfrac{1}{x}\right)^2=x^2+\dfrac{1}{x^2}+2=9$ 이므로

$x+\dfrac{1}{x}=3\ (\because\ x>0)$

$\therefore\ x^3+\dfrac{1}{x^3}=\left(x+\dfrac{1}{x}\right)^3-3\left(x+\dfrac{1}{x}\right)$

$=3^3-3\times 3=18$

23 정답 $112\sqrt{3}$

$x^2-4x+1=0$ 에서 $x>1$ 이므로 양변을 x 로 나누면

$x+\dfrac{1}{x}=4$

$\left(x-\dfrac{1}{x}\right)^2=\left(x+\dfrac{1}{x}\right)^2-4=12$ 이므로

$x-\dfrac{1}{x}=2\sqrt{3}\ (\because\ x>1)$

$x^2+\dfrac{1}{x^2}=\left(x+\dfrac{1}{x}\right)^2-2=14$ 이므로

$x^4-\dfrac{1}{x^4}$

$=\left(x^2+\dfrac{1}{x^2}\right)\left(x^2-\dfrac{1}{x^2}\right)$

$=\left(x^2+\dfrac{1}{x^2}\right)\left(x+\dfrac{1}{x}\right)\left(x-\dfrac{1}{x}\right)$

$=14\times 4\times 2\sqrt{3}$

$=112\sqrt{3}$

해설 SOLUTIONS

24 정답 7

$\frac{y+z}{x}=\frac{z+x}{y}=\frac{x+y}{z}=k$ 라 하면

$y+z=kx,\ z+x=ky,\ x+y=kz$ ····· ㉠

세 식을 변끼리 더하면

$2(x+y+z)=k(x+y+z)$

$(k-2)(x+y+z)=0$

$\therefore\ k=2$ 또는 $x+y+z=0$

(i) $k=2$ 일 때

㉠에서 $y+z=2x,\ z+x=2y,\ x+y=2z$ 이므로

$\frac{(x+y)(y+z)(z+x)}{xyz}=\frac{2z\times 2x\times 2y}{xyz}=8$

$\therefore\ p=8$

(ii) $x+y+z=0$ 일 때

$x+y=-z,\ y+z=-x,\ z+x=-y$ 이므로

$\frac{(x+y)(y+z)(z+x)}{xyz}=\frac{(-z)\times(-x)\times(-y)}{xyz}=-1$

$\therefore\ p=-1$

(i), (ii)에 의하여 가능한 모든 실수 p 의 값의 합은

$8+(-1)=7$